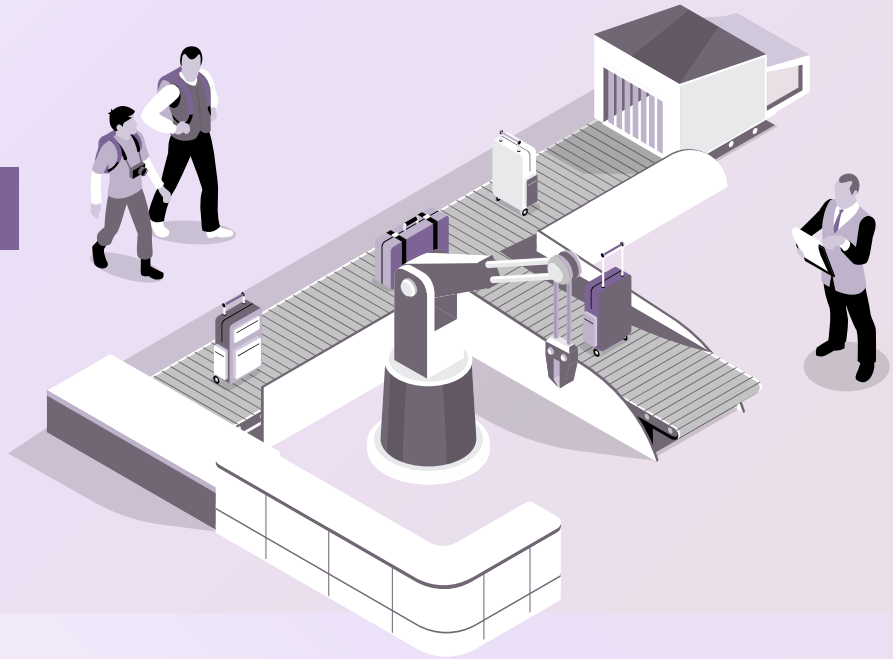
AIRPORT NETWORK

Network Design Case Study

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INTRODUCTION

THE CHALLENGE

An airport has multiple departments that require reliable communication and secure, organized network access.

- Administration
- Check-in & Ticketing
- Security & Baggage Handling
- Operations & IT

OUR APPROACH

Designed and implemented a network using Cisco Packet Tracer with:

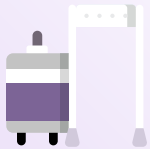
- Routers and Switches
- VLANs for Departmental Separation
- DHCP for Automatic IP Addressing
- OSPF for Routing
- Wireless Access Points for Connectivity

REQUIREMENTS VS DESIGN



REQUIREMENTS

Secure department separation, shared services, Wi-Fi and managed IP addressing.



OUR DESIGN

9 VLANs(adm. Checkin, baggage, security etc.,) , router-on-a-stick(comm between VLANs), DHCP on R1(IP addressing), OSPF to R2-Edge, shared servers in VLAN 90, WPA2 for Wi-Fi.

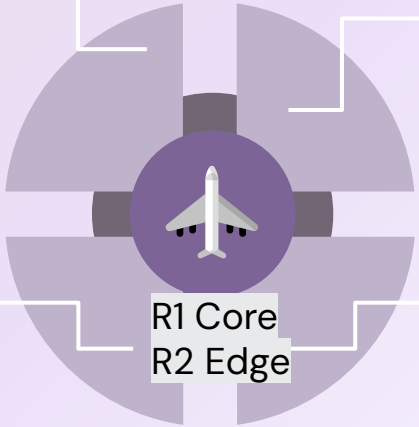
Network Topology by Department

ADMINISTRATION

Admin and check-in PCs connected via SW1-Admin

OPERATIONS & SECURITY

Security and baggage PCs connected via SW2-Operations



IT & SERVICES

IT/Ops PCs, provides DNS, web and airport servers via SW3-Services

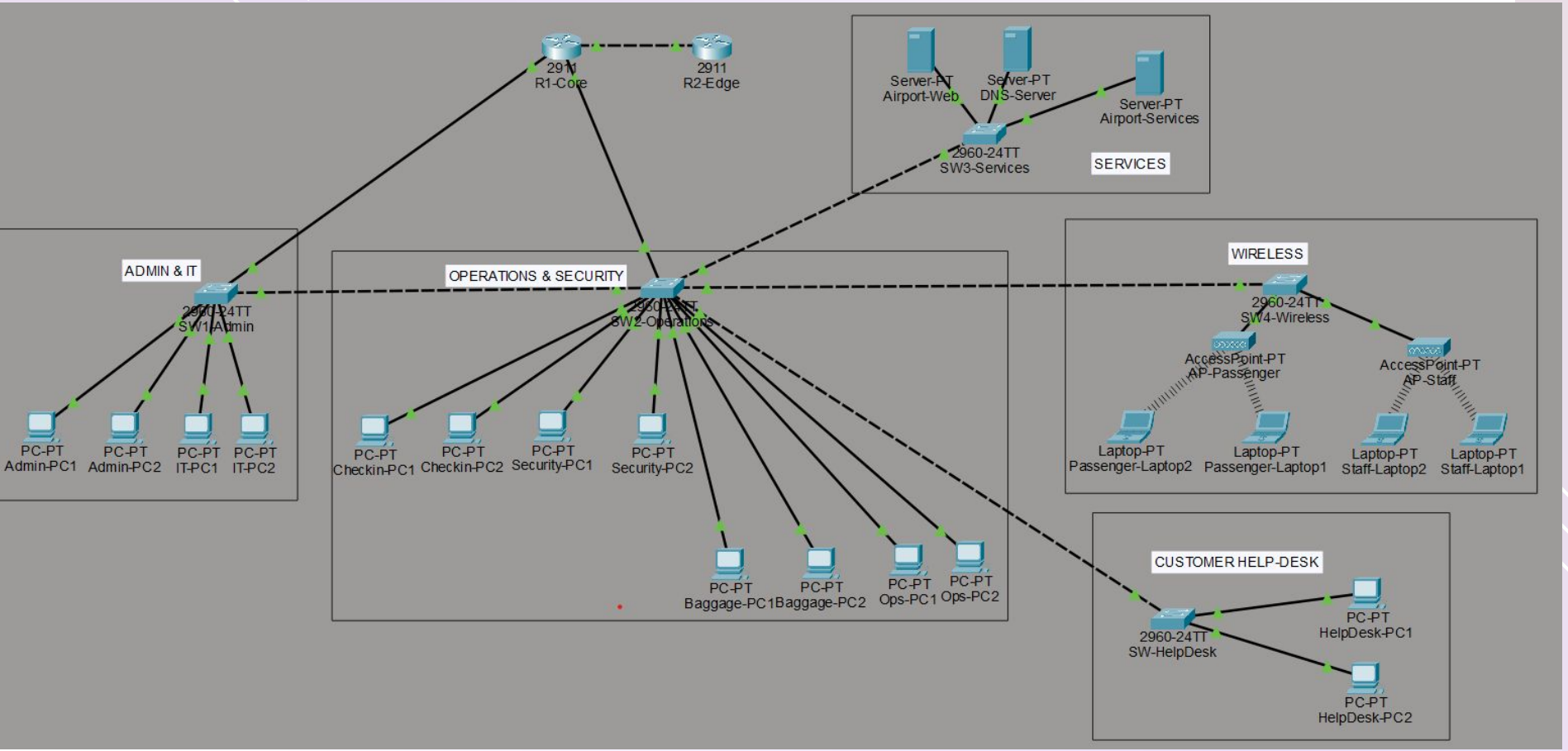
WIRELESS

Passenger and staff Wi-Fi via SW4-Wireless and 2 Access points

R1 handles inter-VLAN routing and DHCP

R2 connects to core router learns int. network

NETWORK TOPOLOGY



DESIGN APPROACH

VLANS

9 VLANs: Admin, Check-in, Security, Baggage, Ops, IT, Wi-Fi x2, Servers

ROUTING

OSPF between R1-Core and R2-Edge over 10.0.0.0/30

INTER-VLAN Routing

R1-Core 802.1Q subinterfaces (router-on-a-stick) communication through logical subinterfaces.

SERVICES

DNS, Web, FTP and Email hosted in server VLAN 90

ADDRESSING

One /24 per VLAN. DHCP for clients, static IPs for servers

WIRELESS

Separated Staff (VLAN 70) and Passenger (VLAN 80) keep traffic logically separated

VLAN AND IP ADDRESSING

VLAN	Department	Subnet	Gateway (R1-Core)	Access switch
10	Administration	192.168.10.0/24	192.168.10.1	SW1-Admin
20	Check-in & Ticketing	192.168.20.0/24	192.168.20.1	SW1-Admin
30	Security	192.168.30.0/24	192.168.30.1	SW2-Operations
40	Baggage Handling	192.168.40.0/24	192.168.40.1	SW2-Operations
50	Airport Operations	192.168.50.0/24	192.168.50.1	SW3-Services
60	IT Department	192.168.60.0/24	192.168.60.1	SW3-Services
70	Staff Wi-Fi	192.168.70.0/24	192.168.70.1	SW4-Wireless
80	Passenger Wi-Fi	192.168.80.0/24	192.168.80.1	SW4-Wireless
90	Servers	192.168.90.0/24	192.168.90.1	SW3-Services

- **Router link:** R1 Gi0/0 10.0.0.1/30 – R2 Gi0/0 10.0.0.2/30 .Our two routers are connected through the 10.0.0.0/30 network. R1-Core uses 10.0.0.1 and R2-Edge uses 10.0.0.2. A /30 subnet provides exactly two conventionally usable host addresses, making it suitable for this point-to-point connection.
- **Servers (static):** DNS 192.168.90.10 · Web 192.168.90.20 · Airport-Services 192.168.90.30. We use static addresses so clients can reliably reach these services.
- **DHCP:** first 10 addresses of each client subnet excluded; DNS option 192.168.90.10

SWITCHING: VLANS AND TRUNKS

- **Local VLAN configuration:** We configured VLANs on different switches according to our design. SW1-Admin contains VLANs 10 and 20, SW2-Operations contains VLANs 30 to 90, SW3-Services contains VLANs 50, 60, and 90, and SW4-Wireless contains VLANs 70 and 80.
- **Local access ports:** carry traffic for one VLAN and are generally used for end devices. Hosts attach as untagged access ports, e.g. SW1 Fa0/3–4 belong to VLAN 10, SW2 Fa0/5–6 in VLAN 30 and Fa0/7–8 in VLAN 40, SW3 Fa0/6–8 in VLAN 90
- **Trunk transport:** carry traffic from multiple VLANs over a single physical link. In our project, SW2 FastEthernet 0/1 to 0/4 are configured as trunk ports.We used IEEE 802.1Q, or dot1Q, for VLAN tagging. It adds VLAN identification to Ethernet frames so that the receiving device can identify which VLAN the traffic belongs to.Trunking reduces the need for separate physical connections for every VLAN and makes the network easier to manage.
- So, access ports connect end devices to individual VLANs, while trunk ports carry multiple VLANs between network devices

SW2 trunk ports

```
SW2-Operations(config)#do show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1
Fa0/3       on           802.1q         trunking      1
Fa0/4       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005
Fa0/2       1-1005
Fa0/3       1-1005
Fa0/4       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,30,40,50,60,70,80,90
Fa0/2       1,30,40,50,60,70,80,90
Fa0/3       1,30,40,50,60,70,80,90
Fa0/4       1,30,40,50,60,70,80,90

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,30,40,50,60,70,80,90
Fa0/2       none
Fa0/3       none
  --More--
```

SW3 VLAN assignment

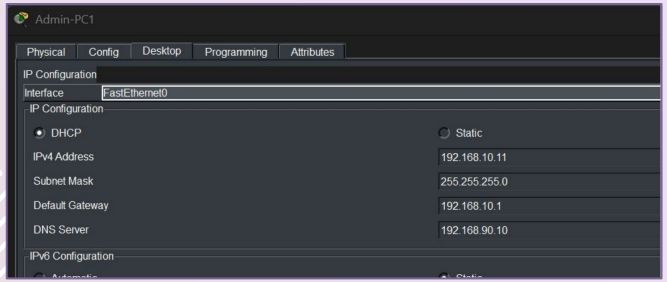
```
SW3-Services(config)#do show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
50   OPERATIONS                       active    Fa0/2, Fa0/3
60   IT                               active    Fa0/4, Fa0/5
90   SERVERS                          active    Fa0/6, Fa0/7, Fa0/8
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW3-Services(config)#
```

INTER-VLAN ROUTING AND DHCP

R1-Core subinterfaces (gateways)

```
Router(config)#do show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.0.1        YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  up                    up
GigabitEthernet0/1.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/1.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  up                    up
GigabitEthernet0/2.30  192.168.30.1    YES manual up                    up
GigabitEthernet0/2.40  192.168.40.1    YES manual up                    up
GigabitEthernet0/2.50  192.168.50.1    YES manual up                    up
GigabitEthernet0/2.60  192.168.60.1    YES manual up                    up
GigabitEthernet0/2.70  192.168.70.1    YES manual up                    up
GigabitEthernet0/2.80  192.168.80.1    YES manual up                    up
GigabitEthernet0/2.90  192.168.90.1    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router(config)#
```



Admin-PC1 receives its address by DHCP

Router-on-a-stick uses logical subinterfaces on a router, with each subinterface associated with a VLAN and assigned its default gateway address.

- For example, our VLAN 10 configuration is:
- interface gigabitEthernet 0/1.10
- encapsulation dot1Q 10
- ip address 192.168.10.1 255.255.255.0

- **DHCP on R1:** We configured DHCP on R1-Core, with separate pools for client VLANs. We also excluded the first ten addresses of each client subnet from DHCP allocation and provided the DNS server address, 192.168.90.10.
- **Outcome:** clients get addresses automatically, e.g. Admin-PC1 192.168.10.11 and Staff-Laptop1 192.168.70.11

To conclude, router-on-a-stick enables communication between VLANs, while DHCP automatically configures client devices. Together, they make our airport network easier to manage and allow devices to communicate across the appropriate networks.

ROUTING: OSPF

- **Point-to-point link:** R1-Core 10.0.0.1/30 and R2-Edge 10.0.0.2/30
- **OSPF** advertises all nine VLAN subnets from R1-Core to R2-Edge
- **R2-Edge** learns 192.168.10.0 to 192.168.90.0 /24 as O routes via 10.0.0.1

show ip route on R2-Edge

```
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
O    192.168.10.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.20.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.30.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.40.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.50.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.60.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.70.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.80.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.90.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
```

9

VLAN routes learned by R2

100%

R1–R2 ping success (5/5)

/30

10.0.0.0 link subnet

SHARED SERVICES



DNS SERVER

192.168.90.10 · DNS (UDP 53). Resolves airport.local, www.airport.local and services.airport.local



WEB SERVER

192.168.90.20 · HTTP (TCP 80). Airport portal opens at http://airport.local



FTP + EMAIL

Airport-Services 192.168.90.30 · FTP (TCP 21) and Email (SMTP/POP3, airport.com)

WIRELESS ACCESS



STAFF WI-FI

SSID Airport-Staff · VLAN 70 · WPA2-PSK (AES) · channel 6. Laptops get 192.168.70.x by DHCP



PASSENGER WI-FI

SSID Airport-Passenger · VLAN 80 · WPA2-PSK (AES) · channel 6. Gateway 192.168.80.1

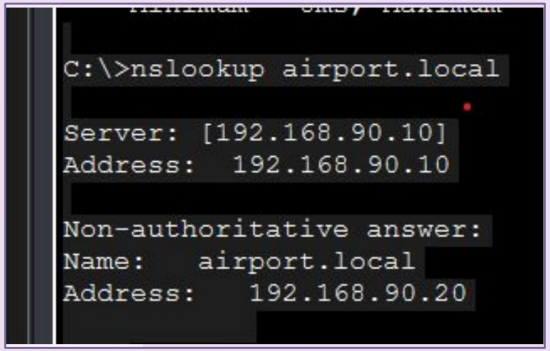
CONNECTIVITY TESTING

Source	VLAN	Gateway	DNS server .90.10	Web / Services
Admin-PC1	10	✓	✓	–
Check-in PC	20	✓	✓	–
Security-PC1	30	✓	✓	✓ ping + browser
Baggage-PC1	40	✓	✓	–
Ops-PC1	50	✓	✓	–
IT-PC1	60	✓	✓	–
Staff-Laptop1	70	✓	✓	✓ web .90.20, services .90.30
Passenger-Laptop1	80	✓	✓	✓ browser
Passenger-Laptop2	80	✓	✓	–

- **Ping results:** All recorded ping tests completed successfully (4/4 replies, 0% packet loss).
- **TTL 127** on server pings shows the packets crossed R1-Core (routed between VLANs)

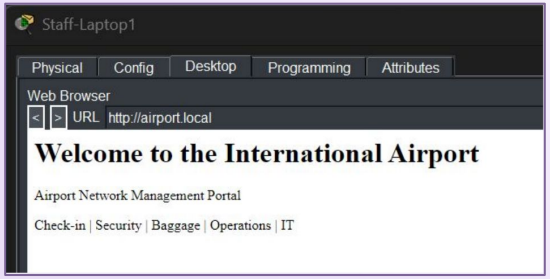
APPLICATION-LAYER SERVICES

DNS (UDP 53)



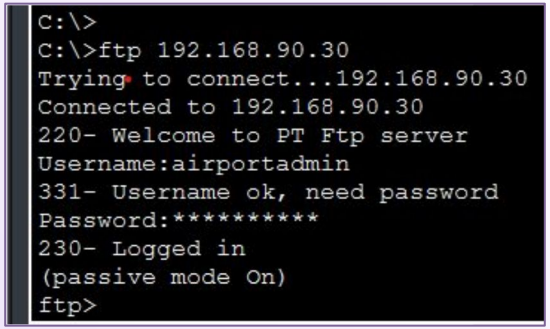
nslookup airport.local returns 192.168.90.20

HTTP (TCP 80)

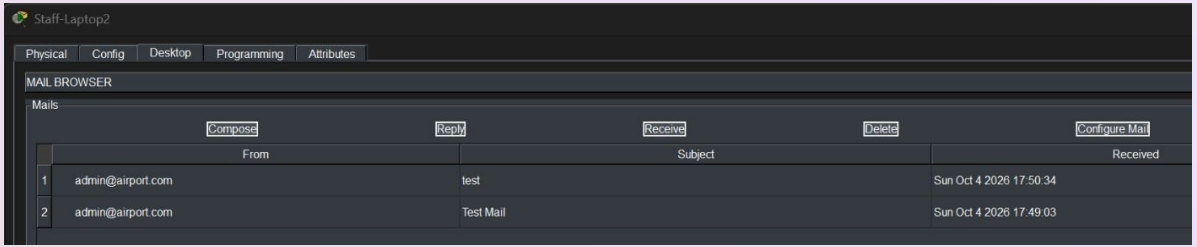


Portal opens by name from staff, security and passenger devices

FTP (TCP 21)



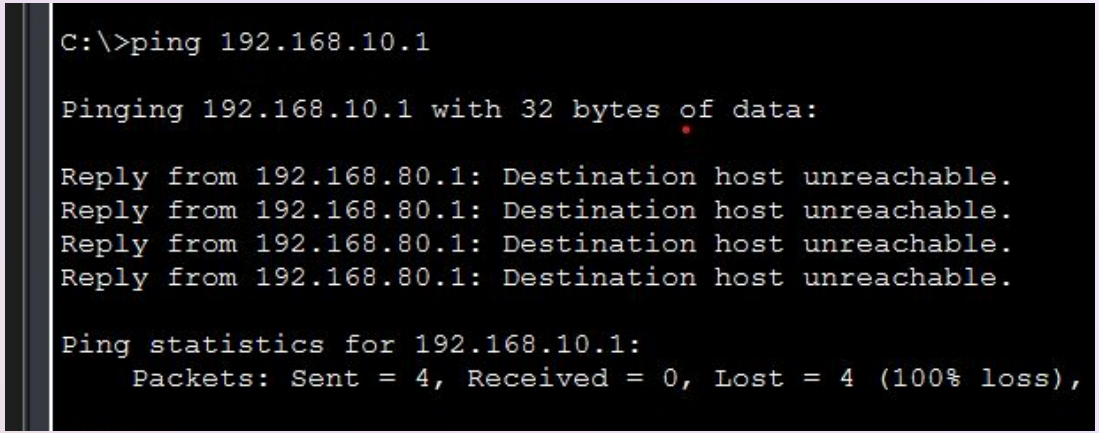
Login to 192.168.90.30 succeeds (230 Logged in)



- **Email:** SMTP and POP3 on Airport-Services, domain airport.com, users admin and ops. Mail sent from admin to ops was delivered

ACL Implementation

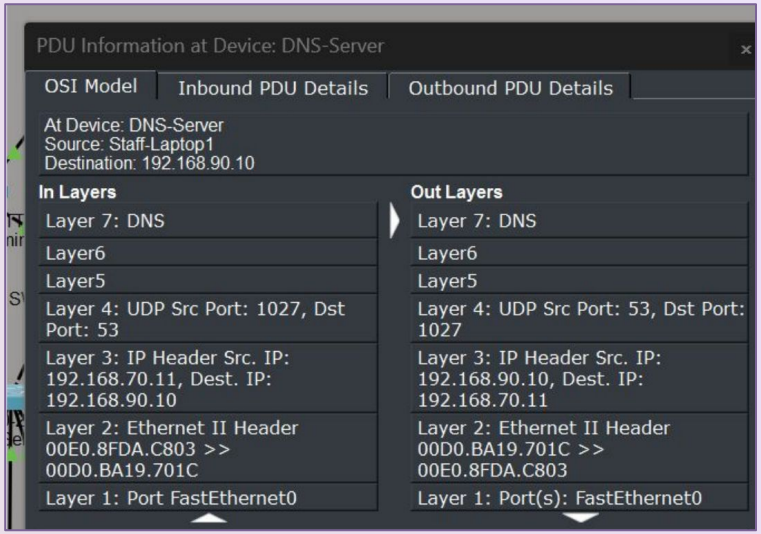
ACL 1 - Stops Passenger Wifi from reaching the internal departments.



Passenger Wi-Fi (192.168.80.12) pinging the Admin gateway (192.168.10.1): blocked by the PASSENGER-RESTRICT ACL.

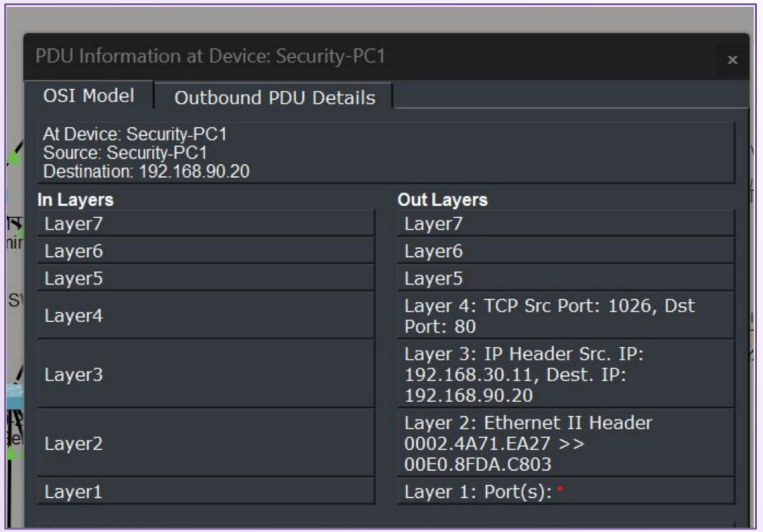
SIMULATION MODE: PACKET TRACE

DNS query (UDP): Staff-Laptop 1 to DNS-Server



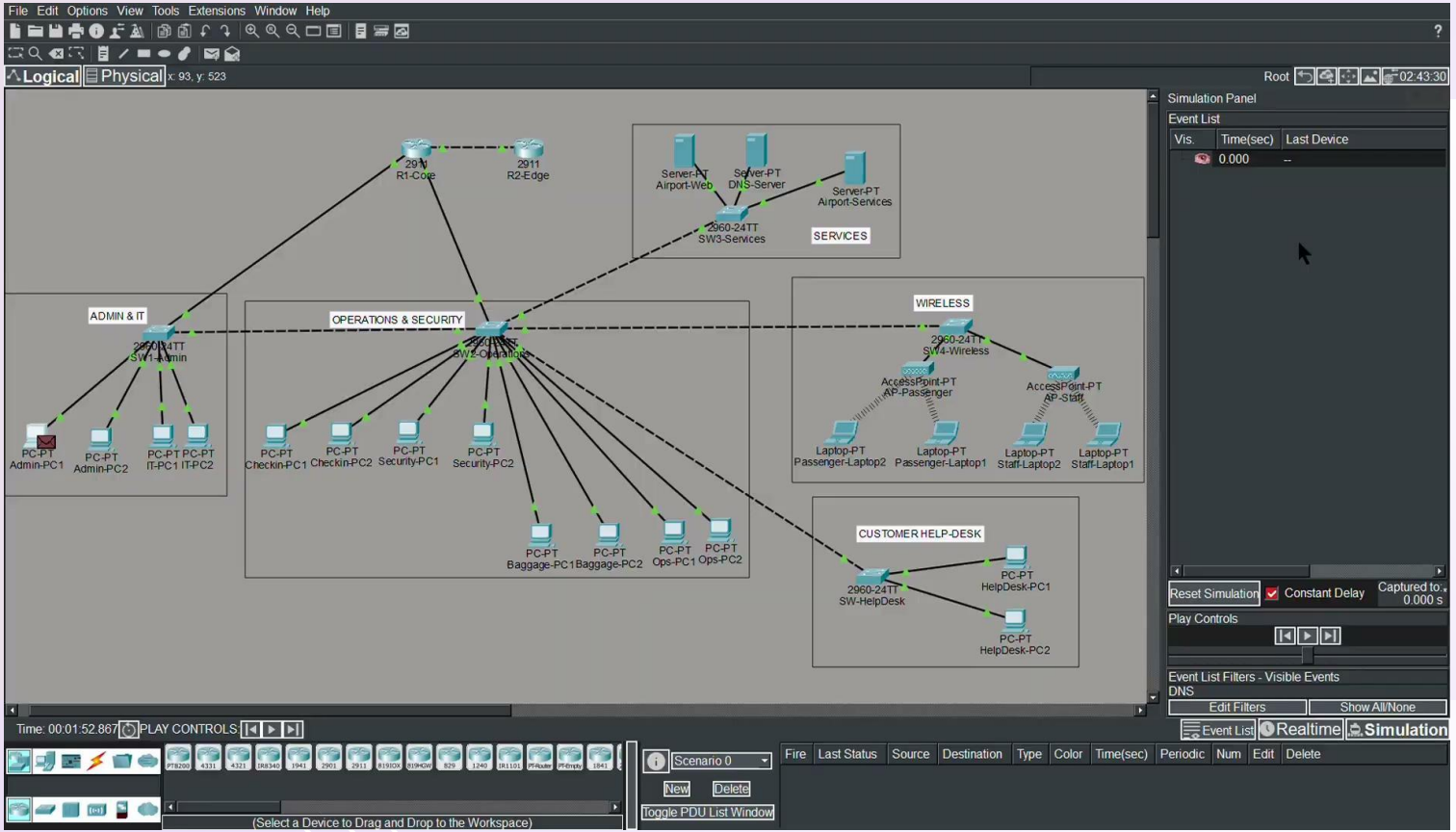
- **App/Transport:** DNS over UDP 1027 to 53
- **Network/Link:** 192.168.70.11 to 192.168.90.10, Ethernet II frame
- **Path:** AP-Staff, SW4, SW2, R1-Core, SW2, SW3, DNS-Server

HTTP (TCP): Security-PC1 to Airport-Web

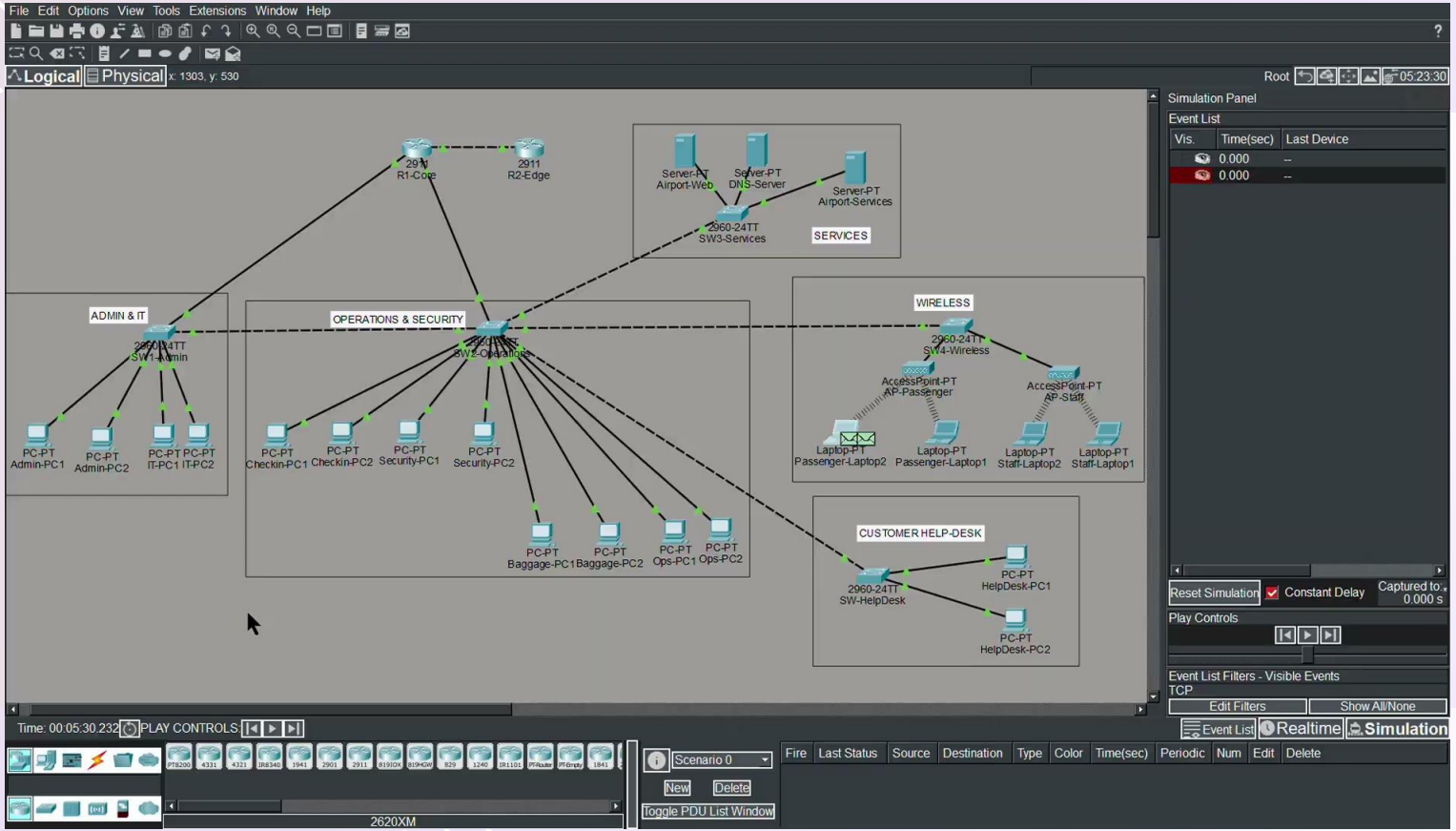


- **App/Transport:** HTTP over TCP 1026 to 80, connection opened then closed (FIN+ACK)
- **Network/Link:** 192.168.30.11 to 192.168.90.20, Ethernet II frame
- **Routing:** R1-Core forwards between VLAN 30 and VLAN 90

DNS Packet Trace Simulation



HTTP TCP Packet Trace Simulation



THANK YOU

